

Student 24 **MALLI CHANDANA** 192472250RD

2 / 2 submitted

Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
interface DeliveryCharge {
    double calculateDeliveryCharge(double foodCost);
}
interface CouponDiscount {
    double calculateDiscount(double foodCost,String couponCode);
}
class FoodOrder implements DeliveryCharge, CouponDiscount{
    public double
        calculateDeliveryCharge(double foodCost){
            if (foodCost >= 500)
                return 0;
            else
                return 50;
        }
    public double calculateDiscount(double foodCost,String couponCode){
        if
            (couponCode.equals("SAVE10"))
                return foodCost * 0.10;
            else
                return 0;
        }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        double foodCost = sc.nextDouble();
        String couponCode = sc.next();
        FoodOrder order = new FoodOrder();
        double deliveryCharge = order.calculateDeliveryCharge(foodCost);
        double discount = order.calculateDiscount(foodCost,couponCode);
        double finalBill = foodCost + deliveryCharge - discount;
        System.out.println("Food Cost : " + foodCost);
        System.out.println("Delivery Charge : " + deliveryCharge);
        System.out.println("Discount : " + discount);
        System.out.println("Final Bill : " + finalBill);
    }
}
```

INPUT

600
SAVE10

OUTPUT

Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Room{
    int roomNumber;
    String guestName;
    double roomRent;
    Room(int roomNumber, String guestName, double roomRent){
        this.roomNumber = roomNumber;
        this.guestName = guestName;
        this.roomRent = roomRent;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int roomNumber = sc.nextInt();
        sc.nextLine();
        String guestName = sc.nextLine();
        double roomRent = sc.nextDouble();
        HotelBill bill = new HotelBill(roomNumber, guestName, roomRent);
        double serviceCharge = bill.calculateServiceCharge();
        double finalBill = roomRent + serviceCharge;
        System.out.println("Room Number : " + roomNumber);
        System.out.println("Guest Name : " + guestName);
        System.out.println("Room Rent : " + roomRent);
        System.out.println("Service Charge : " + serviceCharge);
        System.out.println("Final Bill : " + finalBill);
        sc.close();
    }
}
interface ServiceCharge{
    double calculateServiceCharge();
}
class HotelBill extends Room implements ServiceCharge{
    HotelBill(int roomNumber, String guestName, double roomRent){
        super(roomNumber, guestName, roomRent);
    }
    public double calculateServiceCharge(){
        if(roomRent >= 5000)
            return roomRent * 0.12;
        else
            return roomRent * 0.08;
    }
}
```

INPUT

205
Ananya
6000

OUTPUT

Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0